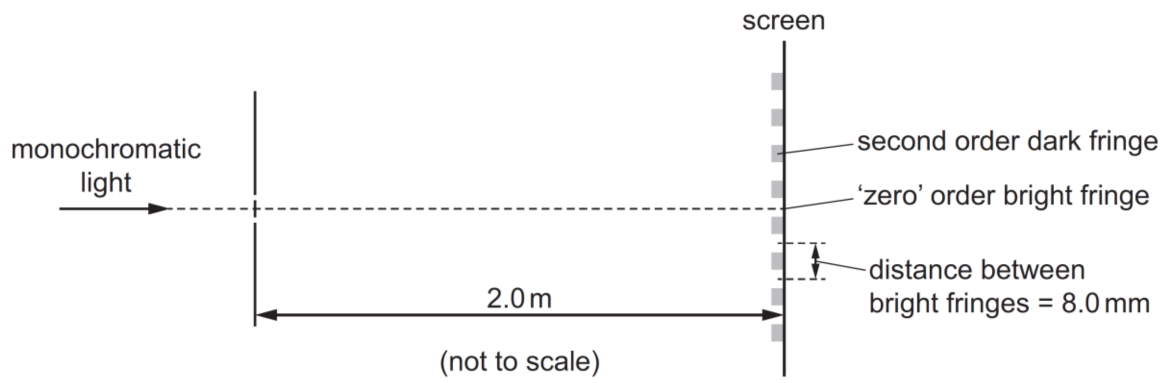
19 Monochromatic light is incident on a pair of narrow slits a distance of 0.1 mm apart. A series of bright and dark fringes are observed on a screen a distance of 2.0 m away. The distance between adjacent bright fringes is 8.0 mm.



What is the path difference between the light waves from the two slits that meet at the second order dark fringe?

A	2.0×10^{-7} m
B	4.0×10^{-7} m
C	6.0×10^{-7} m
D	8.0×10^{-7} m

Answer: C

Double slit equation,

$$x = \frac{\lambda D}{a}$$

$$\lambda = \frac{xa}{D} = \frac{8\times10^{-3} \times 0.1\times10^{-3}}{2} = 4\times10^{-7}$$

At the n = 2 dark fringe, the path difference travelled by the light waves are 1.5 λ
Path difference = $1.5\lambda = 6 \times 10^{-7}$ m

20 Two parallel circular metal plates X and Y, each of diameter 18 cm, have a separation of